

Drudejev model

$$\vec{F} = m\vec{a}$$

$$m\vec{a} = \vec{F}_e - \frac{m\vec{v}}{\tau} = e\vec{E} - \frac{m\vec{v}}{\tau} \rightarrow \vec{j} = \sigma\vec{E}$$

$$\vec{E}(t) = \vec{E}_0 \cos \omega t = \vec{E}_0 e^{-i\omega t}$$

$$\vec{j}(t) = ?$$

$$m\vec{a} = \vec{F}_e - \frac{m\vec{v}}{\tau} = e\vec{E}_0 e^{-i\omega t} - \frac{m\vec{v}}{\tau}, \quad \vec{j} = ne\vec{v}$$

$$\frac{m}{ne}\vec{j} = e\vec{E}_0 e^{-i\omega t} - \frac{m}{ne}\vec{j} \quad \vec{j} = \vec{j}_0 e^{-i\omega t}$$

$$\frac{m}{ne}\vec{j}_0 = e\vec{E}_0 - \frac{m}{ne}\vec{j}_0$$

$$\vec{j}_0 \left(\frac{m}{ne} + \frac{m}{ne} \right) = e\vec{E}_0$$

$$\vec{j}_0 = \frac{e}{\frac{m}{ne} + \frac{m}{ne}} \vec{E}_0 = \frac{ne^2}{m} \left(\frac{1}{\tau} + i\omega \right)^{-1} \vec{E}_0 = \frac{ne^2}{m} \frac{\tau}{1 - i\omega\tau} \vec{E}_0 = \underbrace{\sigma(\omega)}_{\sigma(\omega)} \vec{E}_0$$

$$\omega\tau \ll 1 \rightarrow \sigma(\omega) = \sigma(0)$$

$$\omega\tau \gg 1 \rightarrow \sigma(\omega) = \frac{ine^2}{m\omega}$$

Približek $\sigma = \text{konst.}$

$V = 0$ (postavimo V na 0)

$$H = \frac{p^2}{2m}$$

$$\mathcal{E}(\vec{k}) = \frac{\hbar^2 k^2}{2m} \rightarrow d\mathcal{E} = \frac{\hbar^2}{2m} k dk$$

$$\psi(x+L) = \psi(x)$$

$$3D: V_E = \frac{8\pi^3}{L^3} = \frac{8\pi^3}{V}$$

$$\psi = \frac{1}{\sqrt{V}} e^{i\vec{k}\cdot\vec{r}}$$

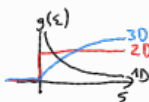
$$2D: V_E = \frac{4\pi^2}{S}$$

$$e^{i\vec{k}\cdot\vec{L}} = 1$$

$$\vec{k} = \frac{2\pi}{L} n, \quad n \in \{x, y, z\}$$

$$g(\epsilon) = \frac{1}{V} \frac{dN(\epsilon, \epsilon+d\epsilon)}{d\epsilon} \quad \text{gostota stanj}$$

$$2D: g(\epsilon) = \frac{1}{S} \frac{2\pi k dk}{\frac{\hbar^2 k^2}{2m}} \cdot 2 = \frac{k}{\pi} \frac{dk}{d\epsilon} = \frac{k}{\pi} \frac{m}{\hbar^2 k} = \frac{m}{\pi \hbar^2}$$



$$g(\epsilon) = \begin{cases} \frac{m}{\pi \hbar^2}, & \epsilon \geq 0 \\ 0, & \epsilon < 0 \end{cases}$$

$$= \frac{m}{\pi \hbar^2} \Theta(\epsilon)$$

$$3D: g(\epsilon) = \frac{1}{V} \frac{4\pi k^2 dk}{\frac{8\pi^3}{V}} \cdot 2 = \frac{k^2}{\pi^2} \frac{dk}{d\epsilon} = \frac{k^2}{\pi^2} \frac{m}{\hbar^2 k} = \frac{m}{\pi^2 \hbar^2} k = \frac{m}{\pi^2 \hbar^2} \sqrt{\frac{2m\epsilon}{\hbar^2}}$$

$$1D: g(\epsilon) = \frac{1}{L} \frac{2 dk}{\frac{2\pi \hbar^2 k}{L}} \cdot 2 = \frac{2}{\pi} \frac{dk}{d\epsilon} = \frac{2}{\pi} \frac{m}{\hbar^2 k} = \frac{2m}{\pi \hbar^2} \frac{1}{\sqrt{\frac{2m\epsilon}{\hbar^2}}}$$

$$g(\epsilon) = A \epsilon^{\frac{d}{2}-1} \Theta(\epsilon)$$

$$d=1, 2, 3$$

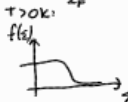
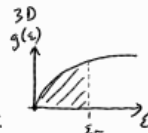
(dimenzija)

$$N = V \int_{-\infty}^{\epsilon_F} g(\epsilon) f(\epsilon) d\epsilon, \quad T=0K: f(\epsilon) = \Theta(\epsilon_F - \epsilon)$$

$$N(T=0K) = V \int_{-\infty}^{\epsilon_F} g(\epsilon) d\epsilon, \quad T>0K: f(\epsilon) = \frac{1}{e^{\beta(\epsilon - \mu)} + 1}$$

$$\mu(T)$$

$$\mu(0K) = \epsilon_F$$



Kako se s T spreminja μ , da bo št. e konst.?

$$N(T=0) = V \int_{-\infty}^{\epsilon_F} g(\epsilon) d\epsilon = N(T>0K) = V \int_{-\infty}^{\infty} g(\epsilon) f(\epsilon) d\epsilon$$

Sommerfeldov razvoj

$$\mu(T) = \epsilon_F - \frac{\pi^2}{6} (k_B T)^2 \frac{g'(\epsilon_F)}{g(\epsilon_F)}$$

$$g(\epsilon) = A \epsilon^{\frac{d}{2}-1} \Theta(\epsilon)$$

$$g'(\epsilon) = A \left(\frac{d}{2}-1 \right) \epsilon^{\frac{d}{2}-2} \Theta(\epsilon)$$

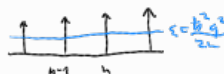
$$\mu(T) = \epsilon_F - \frac{\pi^2}{6} (k_B T)^2 \left(\frac{d}{2}-1 \right) \epsilon_F^{-1} = \epsilon_F \left[1 - \frac{\pi^2}{6} (k_B T)^2 \left(\frac{d}{2}-1 \right) \epsilon_F^{-2} \right]$$

1D Kronig-Penneyev model

$$V(x) = \lambda \sum_n \delta(x - na), \quad \lambda > 0$$

$$H = \frac{p^2}{2m} + V(x)$$

$$q = \sqrt{\frac{2mE}{\hbar^2}}$$



$$\psi_{n-1} = A_{n-1} e^{iqx} + B_{n-1} e^{-iqx}$$

$$\psi_n = A_n e^{iqx} + B_n e^{-iqx}$$

R.P.:

$$\psi_{n-1}(na) = \psi_n(na)$$

$$\psi'_n(na) - \psi'_{n-1}(na) = \frac{2m\lambda}{\hbar^2} \psi_n(na)$$

$$T: \psi(x) \rightarrow \psi(x+na)$$

$$[H, T] = 0$$

$$T^N \psi(x) = \psi(x + Na) = \psi(x)$$

(invarianca na translacijo)

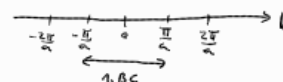
$$T \psi(x) = c \psi(x)$$

$$T^N \psi(x) = c^N \psi(x) = \psi(x)$$

$$c^N = 1$$

$$c = e^{ik_n a}, \quad k_n \in \mathbb{R}$$

$$e^{ik_n a} = 1 \rightarrow k = \frac{2\pi}{Na} z, \quad z \in \mathbb{Z}$$



$T \psi(x) = \psi(x+a) = e^{ika} \psi(x)$

$A_n e^{iq(x+a)} + B_n e^{-iq(x+a)} = e^{ika} (A_{n-1} e^{iqx} + B_{n-1} e^{-iqx}) \rightarrow A_{n-1} = A_n e^{i(q-k)a}, B_{n-1} = B_n e^{-i(q+k)a}$

RP pri $x=na$: $A_{n-1} e^{iqna} + B_{n-1} e^{-iqna} = A_n e^{iqna} + B_n e^{-iqna}$

$i q (A_n e^{iqna} - B_n e^{-iqna}) - i q (A_{n-1} e^{iqna} - B_{n-1} e^{-iqna}) = 2Q (A_n e^{iqna} + B_n e^{-iqna}), Q = \frac{m\lambda}{\hbar^2}$

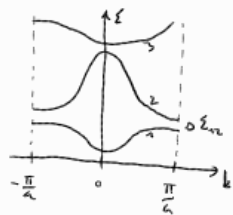
vstavimo u^o A_{n-1}, B_{n-1} . Dobimo sistem 2 enačb, ki ima rešitev, če je $\det=0$.

$\rightarrow \cos ka = \cos qa + Qa \frac{\sin qa}{qa}$

$Q = \frac{m\lambda}{\hbar^2}$

$q = \sqrt{\frac{2mE}{\hbar^2}}$

$T \psi_k(x) = \psi_k(x+a) = e^{ika} \psi_k(x)$



$\Delta E_{n,n+1} = \frac{2\lambda}{a}$ za $Qa \ll 1$

Približek šibkega potenciala (skoraj prosti e⁻)

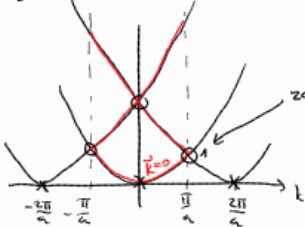
$V(\vec{r} \rightarrow \vec{r}) = V(\vec{r})$

razvijmo Fourierovo vrsto

$V(\vec{r}) = \sum_{\vec{k}} V_{\vec{k}} e^{i\vec{k}\cdot\vec{r}}, V_{\vec{k}} = \frac{1}{V_{\text{cel}}} \int_{\text{oc}} V(\vec{r}) e^{-i\vec{k}\cdot\vec{r}} d\vec{r}$
integral po 1 osh celici

$\psi_{\vec{k}}(\vec{r}) = e^{i\vec{k}\cdot\vec{r}} u_{\vec{k}}(\vec{r}) = \sum_{\vec{k}'} C_{\vec{k}-\vec{k}'} e^{i(\vec{k}-\vec{k}')\cdot\vec{r}}$
 $\hookrightarrow u_{\vec{k}}(\vec{r}) = u_{\vec{k}}(\vec{r} + \vec{R})$

1D:



zaključimo nas razcep v teh točkah

$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = E\psi$

$(\epsilon_{\vec{k}-\vec{k}'} - \epsilon_{\vec{k}}) C_{\vec{k}-\vec{k}'} + \sum_{\vec{k}''} V_{\vec{k}-\vec{k}''} C_{\vec{k}-\vec{k}''} = 0$

popravke delamo samo v stičiščih parabol

hpr. 1:

$\vec{k}_1: (\epsilon_{\vec{k}-\vec{k}_1} - \epsilon_{\vec{k}}) C_{\vec{k}-\vec{k}_1} + \sum_{\vec{k}'' \in \{\vec{k}_1, \vec{k}_2\}} V_{\vec{k}-\vec{k}''} C_{\vec{k}-\vec{k}''} = 0$

$\vec{k}_2: (\epsilon_{\vec{k}-\vec{k}_2} - \epsilon_{\vec{k}}) C_{\vec{k}-\vec{k}_2} + \sum_{\vec{k}'' \in \{\vec{k}_1, \vec{k}_2\}} V_{\vec{k}-\vec{k}''} C_{\vec{k}-\vec{k}''} = 0$

$$\begin{bmatrix} \epsilon_0 - \epsilon + V_0 & V_{\vec{k}_1 - \vec{k}_1} \\ V_{\vec{k}_1 - \vec{k}_2} & \epsilon_0 - \epsilon + V_0 \end{bmatrix} \begin{bmatrix} C_{\vec{k}-\vec{k}_1} \\ C_{\vec{k}-\vec{k}_2} \end{bmatrix} = 0$$

$\det(\dots) = 0$

$(\epsilon_0 - \epsilon + V_0)^2 - \underbrace{V_{\vec{k}_1 - \vec{k}_1} V_{\vec{k}_1 - \vec{k}_2}}_{|V_{\vec{k}_1 - \vec{k}_1}|^2} = 0$

$\epsilon_0 - \epsilon + V_0 = \pm |V_{\vec{k}_1 - \vec{k}_2}|$

$\epsilon_{\pm} = \epsilon_0 + V_0 \mp |V_{\vec{k}_1 - \vec{k}_2}|$

$\Delta \epsilon = \epsilon_+ - \epsilon_- = 2 |V_{\vec{k}_1 - \vec{k}_2}|$

$V_k = \frac{1}{V_{\text{oc}}} \int_{\text{oc}} V(x) e^{-ikx} dx = \frac{1}{a} \int_{-a/2}^{a/2} V(x) e^{-ikx} dx = \frac{1}{a} \int_0^a \lambda \delta(x) e^{-ikx} dx = \frac{1}{2} \lambda$

$\Delta \epsilon = 2 |V_k| = \frac{2\lambda}{a}$

če želimo dobiti razcep energije za drugacen potencial, moramo samo rešiti zadnji integral.

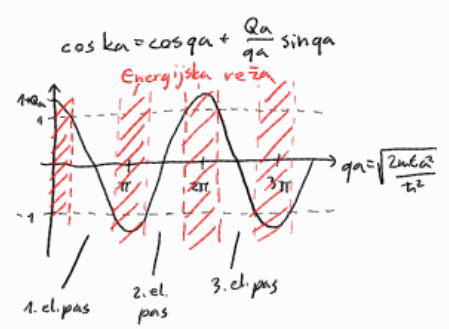
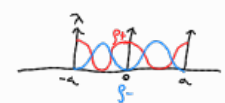
$\psi_{\pm}(x) = \sum_{k \in \{k_1, k_2\}} C_{k \pm} e^{i(k \pm)x}$

$k_1=0, k_2=\frac{2\pi}{a}, k=\frac{\pi}{a} \quad \Sigma_+ \rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\psi_{\pm}(x) = C_{\frac{\pi}{2}}^+ e^{i\frac{\pi}{2}x} + C_{-\frac{\pi}{2}}^+ e^{-i\frac{\pi}{2}x} \quad \Sigma_- \rightarrow \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\psi_+ \propto e^{i\frac{\pi}{2}x} + e^{-i\frac{\pi}{2}x} \propto \cos \frac{\pi}{2}x \rightarrow \Sigma_+ = \epsilon_0 + \frac{2\lambda}{a}, \psi_+ \propto \cos^2 \frac{\pi}{2}x$

$\psi_- \propto e^{i\frac{\pi}{2}x} - e^{-i\frac{\pi}{2}x} \propto \sin \frac{\pi}{2}x \rightarrow \Sigma_- = \epsilon_0, \psi_- \propto \sin^2 \frac{\pi}{2}x$



fcc, periodičan potencijal

reciprocna mreža = bcc

Kako se razloži energije u točki w (točka u ugljenu oktaedru)?

$$W = \frac{4\pi}{a} \left(\frac{1}{2}, 0, \frac{1}{4} \right)$$

$$\begin{bmatrix} \xi^0 - \xi & V_{\vec{k}_1 - \vec{k}_4} & V_{\vec{k}_3 - \vec{k}_1} & V_{\vec{k}_1 - \vec{k}_4} \\ V_{\vec{k}_1 - \vec{k}_2} & \xi^0 - \xi & V_{\vec{k}_1 - \vec{k}_2} & V_{\vec{k}_4 - \vec{k}_2} \\ V_{\vec{k}_1 - \vec{k}_3} & V_{\vec{k}_2 - \vec{k}_3} & \xi^0 - \xi & V_{\vec{k}_1 - \vec{k}_3} \\ V_{\vec{k}_1 - \vec{k}_4} & V_{\vec{k}_2 - \vec{k}_4} & V_{\vec{k}_3 - \vec{k}_4} & \xi^0 - \xi \end{bmatrix} \rightarrow \det(\dots) = 0$$

$$V_0 = \frac{1}{V_{\text{c.k.}}} \int V(\vec{r}) d\vec{r} \rightarrow 0 \text{ (izabrano energijsko skalo tako, da to je 0)}$$

$$V(R\vec{r}) = V(\vec{r})$$

↑
simetrijska operacija

$$V_{\vec{k}} = V_{-\vec{k}}, \quad V_{\vec{k}} = V_{\vec{k}}, \quad V_{R\vec{k}} = V_{\vec{k}}$$

$$V_{\vec{k}_1 - \vec{k}_4} = V_{\vec{k}_1 - \vec{k}_3} = V_1 \in \mathbb{R} \text{ (4-sterana os)}$$

$$V_{\vec{k}_3 - \vec{k}_4} = V_{\vec{k}_4 - \vec{k}_1} = V_{\vec{k}_3 - \vec{k}_2} = V_{\vec{k}_1 - \vec{k}_2} = V_2 \in \mathbb{R} \text{ (zrcaljenje)}$$

$$\det \begin{bmatrix} \xi^0 - \xi & V_1 & V_2 & V_2 \\ V_1 & \xi^0 - \xi & V_2 & V_2 \\ V_2 & V_2 & \xi^0 - \xi & V_1 \\ V_2 & V_2 & V_1 & \xi^0 - \xi \end{bmatrix} = 0$$

$$\rightarrow (\xi^0 - \xi - V_1)^2 (\xi^0 - \xi + V_1)^2 - (2V_2)^2 = 0$$

$$E_{1,2} = \xi^0 - V_1$$

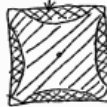
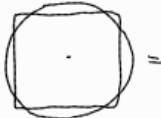
$$E_3 = \xi^0 + V_1 - 2V_2$$

$$E_4 = \xi^0 + V_1 + 2V_2$$

2. KOLOKVIJ

kvadratna mreža

$$V(\vec{r}) = 0$$



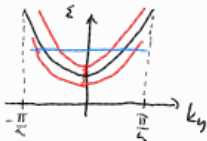
dvajao zasjedena stanja

$$\det \begin{bmatrix} \xi^0 - \xi & V \\ V^* & \xi^0 - \xi \end{bmatrix} = 0 \rightarrow (\xi^0 - \xi)^2 - |V|^2 = 0$$

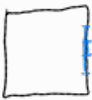
$$\xi^0 - \xi_1 + |V| = 0 \quad \xi_1 = \xi^0 + |V|$$

$$\xi^0 - \xi_2 - |V| = 0 \quad \xi_2 = \xi^0 - |V|$$

$$k_x = \frac{\pi}{a}$$



$$V(\vec{r}) \neq 0$$



$$\begin{vmatrix} \xi^0(\vec{k} - \vec{k}_1) - \xi & V \\ V^* & \xi^0(\vec{k} - \vec{k}_2) - \xi \end{vmatrix} = 0$$

$$(\xi^0(\vec{k} - \vec{k}_1) - \xi)(\xi^0(\vec{k} - \vec{k}_2) - \xi) = |V|^2$$

$$\xi^2 - \xi(\xi^0(\vec{k} - \vec{k}_1) + \xi^0(\vec{k} - \vec{k}_2)) + \xi^0(\vec{k} - \vec{k}_1)\xi^0(\vec{k} - \vec{k}_2) - |V|^2 = 0$$

$$\xi_{\pm} = \frac{\xi^0(\vec{k} - \vec{k}_1) + \xi^0(\vec{k} - \vec{k}_2) \pm \sqrt{(\xi^0(\vec{k} - \vec{k}_1) - \xi^0(\vec{k} - \vec{k}_2))^2 + 4(\xi^0(\vec{k} - \vec{k}_1)\xi^0(\vec{k} - \vec{k}_2) - |V|^2)}}{2}$$

$$= \frac{\xi^0(\vec{k} - \vec{k}_1) + \xi^0(\vec{k} - \vec{k}_2)}{2} \pm \sqrt{(\xi^0(\vec{k} - \vec{k}_1) - \xi^0(\vec{k} - \vec{k}_2))^2 + 4|V|^2}$$

$$\rightarrow \vec{k} = \left(\frac{\pi}{a} - p_x, k_y \right)$$

$$\vec{k}_1 = 0$$

$$\vec{k}_2 = \left(\frac{\pi}{a}, 0 \right)$$

$$\xi_{\pm} = \frac{\hbar^2}{2m} \left(k_y^2 + p_x^2 + \left(\frac{\pi}{a} \right)^2 \right) \pm \sqrt{\frac{\hbar^4}{4m^2} \frac{p_x^2 \pi^2}{a^2} + |V|^2}$$

$$= \frac{\hbar^2}{2m} \left(k_y^2 + p_x^2 + \left(\frac{\pi}{a} \right)^2 \right) \pm |V| \sqrt{1 + \frac{1}{2} \left(\frac{\hbar}{2m} \frac{p_x \pi}{a} \frac{1}{|V|} \right)^2}$$

$$= \frac{\hbar^2}{2m} \left(\left(\frac{\pi}{a} \right)^2 + k_y^2 \right) \pm |V| + \frac{\hbar^2}{2m} p_x^2 \pm \frac{1}{2|V|} \left(\frac{\hbar^2}{2m} p_x \frac{\pi}{a} \right)^2$$

$$= \frac{\hbar^2}{2m} \left(\frac{\pi}{a} \right)^2 \pm |V| + \frac{\hbar^2}{2m} k_y^2 \pm \frac{\hbar^4}{8|V|m^2} \left(\frac{\pi}{a} \right)^2$$

+



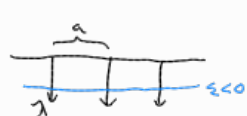
→ dobimo elipse, ne kružnice

-



u vak. smeri ∈ padu, u nasp. pm mreže

Približek tesne vezi

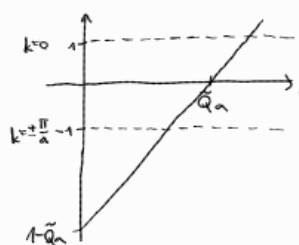


$$\tilde{Q} = \frac{n\lambda}{\hbar^2}, \tilde{Q}a \gg 1$$

$$q = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} = i\tilde{q}$$

$$A_k e^{ikx} + B_k e^{-ikx}$$

$$\rightarrow \cos ka = \cos i\tilde{q}a - \frac{\tilde{Q}a}{i\tilde{q}a} \sin i\tilde{q}a = \cosh \tilde{q}a - \frac{\tilde{Q}a}{i\tilde{q}a} \sinh \tilde{q}a = \cosh(\tilde{q}a) - \frac{\tilde{Q}a}{\tilde{q}a} \sinh(\tilde{q}a)$$



$$\tilde{Q} = -Q = \frac{n\lambda}{\hbar^2}$$

$$\tilde{q}a \rightarrow 0: \cosh \tilde{q}a - \frac{\tilde{Q}a}{\tilde{q}a} \sinh \tilde{q}a \rightarrow 1 - \tilde{Q}a$$

$$\tilde{q}a \rightarrow \infty: \dots \rightarrow \infty$$

$$\xi = \frac{\tilde{q}^2 a^2}{2m}$$

$$\tilde{q}a = \tilde{Q}a + \xi, \xi \ll \tilde{Q}a$$

$$\rightarrow \tilde{q}a = \tilde{Q}a(1 + 2e^{-Qa} \cos ka)$$

$$\psi_0(x) = \sqrt{a} e^{-Q|x|}$$

$$\xi(E) = E_0 - 2\gamma \cos ka$$

$$\gamma = \frac{\hbar^2 Q^2}{m} e^{-Qa}$$

$$\xi(E) = E_0 - \frac{\beta + \sum_{\vec{R}} \gamma(\vec{R}) e^{i\vec{k} \cdot \vec{R}}}{1 + \sum_{\vec{R}} \alpha(\vec{R}) e^{i\vec{k} \cdot \vec{R}}}$$

$$\beta = -\int d^3r \psi_0^*(\vec{r}) \Delta V(\vec{r}) \psi_0(\vec{r})$$

$$\rightarrow \beta = \int d^3r \sqrt{a} e^{-Q|\vec{r}|} \lambda \sum_{n=0}^{\infty} \delta(\vec{r} - n\vec{a}) \sqrt{a} e^{-Q|\vec{r}|} = 2\lambda Q e^{-2Qa}$$

↳ največ prispeva $n=1$, zato faktor 2

$$\gamma(\vec{R}) = -\int d^3r \psi_0^*(\vec{r}) \Delta V(\vec{r}) \psi_0(\vec{r} - \vec{R})$$

$$\gamma = \int d^3r \sqrt{a} e^{-Q|\vec{r}|} \lambda \sum_{n=0}^{\infty} \delta(\vec{r} - n\vec{a}) \sqrt{a} e^{-Q|\vec{r} - n\vec{a}|} = \lambda Q e^{-Qa}$$

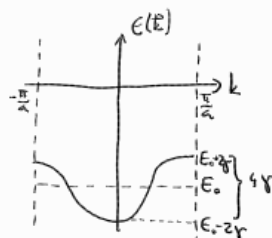
↳ največ prispeva $n=\pm 1$, $k=0$

$$\alpha(\vec{R}) = \int d^3r \psi_0^*(\vec{r}) \psi_0(\vec{r} - \vec{R})$$

$$\alpha \sim Qa e^{-Qa}$$

$$E_0 = -\frac{\hbar^2 Q^2}{2m}$$

$$\rightarrow \xi(E) = E_0 - \frac{\lambda Qa e^{-Qa} (e^{ika} + e^{-ika})}{1 + Qa e^{-Qa}} = E_0 - 2\lambda Q e^{-Qa} \cos ka = E_0 - 2 \underbrace{\frac{Q^2 \hbar^2}{m} e^{-Qa}}_{\gamma} \cos ka$$



↳ postavimo na 0

$$\xi(E) = E_0 - \sum_{\vec{R} \in \text{1. sosed.}} \gamma(\vec{R}) e^{i\vec{k} \cdot \vec{R}} = -2\gamma (\cos k_x a + \cos k_y a)$$

$$\gamma(\vec{R}) \in \mathbb{R}$$

$$\gamma(\vec{a}_1) = \gamma(\vec{a}_2) = \gamma(-\vec{a}_1) = \gamma(-\vec{a}_2) = \gamma$$



$$\xi(\vec{k}) = -\sum_{\vec{R} \in \text{1. s.}} \gamma(\vec{R}) e^{i\vec{k} \cdot \vec{R}} = -\sum_{\vec{R} \in \text{1. s.}} \gamma(\vec{R}) e^{i\vec{k} \cdot \vec{R}} \frac{\vec{R} \cdot \vec{R}}{|\vec{R}|^2} = -\sum_{\vec{R} \in \text{1. s.}} \gamma(\vec{R}) e^{i\vec{k} \cdot \vec{R}} = -\sum_{\vec{R} \in \text{1. s.}} \gamma(\vec{R}) e^{i\vec{k} \cdot \vec{R}} = -\sum_{\vec{R} \in \text{1. s.}} \gamma(\vec{R}) e^{i\vec{k} \cdot \vec{R}} = \xi(\vec{k})$$

$$\Gamma \rightarrow \chi: \vec{k} = (k_x, 0), k_x \in [0, \frac{\pi}{a}]$$

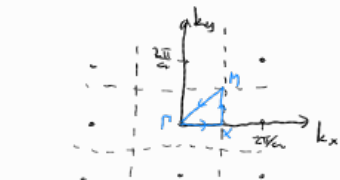
$$\xi(E) = -2\gamma (\cos k_x a + 1)$$

$$\chi \rightarrow \Pi: \vec{k} = (\frac{\pi}{a}, k_y), k_y \in [0, \frac{\pi}{a}]$$

$$\xi(k_y) = -2\gamma (-1 + \cos k_y a)$$

$$\Pi \rightarrow \Gamma: \vec{k} = (k_x, k_y), k_x \in [\frac{\pi}{a}, 0]$$

$$\xi(k_x) = -2\gamma (2 \cos k_x a)$$



⇒ Γ... minimum
Π... maksimum
χ... sedlo

$$\begin{aligned} \Gamma: \xi(k \ll \frac{\pi}{a}) &= -2\gamma (1 - \frac{1}{2}(k_x a)^2 + 1 - \frac{1}{2}(k_y a)^2) = \\ &= -4\gamma + \gamma(k_x^2 + k_y^2)a^2 = -4\gamma + \gamma a^2 k^2 = \\ &= -4\gamma + \frac{\hbar^2 k^2}{2m} \rightarrow m^* = \frac{\hbar^2}{2\gamma a^2} \end{aligned}$$

⇒ min ✓

$$\chi: \vec{k} = (\frac{\pi}{a}, 0) + \vec{q}$$

$$\xi(q \ll \frac{\pi}{a}) = -2\gamma (\cos(\pi + qa) + \cos(qa)) =$$

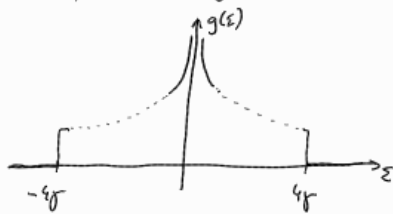
$$= -4\gamma + \gamma(q_x^2 - q_y^2)$$

$$\rightarrow m^* = \begin{bmatrix} -\frac{\hbar^2}{2\gamma a^2} & 0 \\ 0 & \frac{\hbar^2}{2\gamma a^2} \end{bmatrix} \Rightarrow \text{sedlo} \checkmark$$

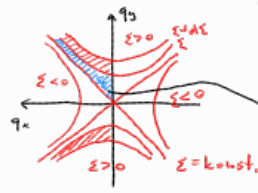
$$\xi(\frac{\pi}{a} - k_x, \frac{\pi}{a} - k_y) = 2\gamma (\cos k_x a + \cos k_y a) = -\xi(E)$$

Γ se spreminja v Π ⇒ ξ(Γ) = -ξ(Π) ⇒ maksimum v Π ✓

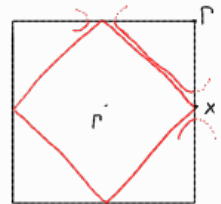
$$\xi(k_x, \frac{\pi}{a} - k_x) = -2\gamma(\cos k_x a - \cos k_x a) = 0$$



točka X:



$$g(\varepsilon) = \frac{1}{V} \frac{dN(\varepsilon, \varepsilon + d\varepsilon)}{d\varepsilon}$$



$$G(\varepsilon) = \frac{1}{V} N(0, \varepsilon)$$

$$g(\varepsilon) = G'(\varepsilon)$$

$$q_x = \sqrt{q_x^2 + \frac{\varepsilon}{\gamma a^2}}$$

$$N(0, \varepsilon) = 2 \frac{\xi_F(q_x)}{\left(\frac{\pi}{L}\right)^2} \cdot 8 \rightarrow \text{ker je v 1. Brill. coni 8 točk, polj}$$

$$\xi_F(q_x) = \int_0^{q_x} dq_x \left[\sqrt{q_x^2 + \frac{\varepsilon}{\gamma a^2}} - \sqrt{q_x^2 - 0} \right]$$

$$G(\varepsilon) = \frac{16}{V} \left(\frac{L}{2\pi} \right)^2 \int_0^{q_x} dq_x \left[\sqrt{q_x^2 + \frac{\varepsilon}{\gamma a^2}} - q_x \right]$$

$$g(\varepsilon) = G'(\varepsilon) = \frac{16}{V} \left(\frac{L}{2\pi} \right)^2 \int_0^{q_x} dq_x \frac{1}{2\gamma a^2} \left(q_x^2 + \frac{\varepsilon}{\gamma a^2} \right)^{-1/2} - \frac{16}{2\gamma a^2} \left(\frac{L}{2\pi} \right)^2 \text{arsh} \frac{q_x a \sqrt{\gamma}}{\sqrt{\varepsilon}}$$

$$\rightarrow \text{arsh } x \approx y \quad \varepsilon \text{ majhen} \rightarrow \text{arsh} \dots \gg 1$$

$$x = \text{sh } y \approx \frac{e^y}{2}$$

$$y \approx \log 2x$$

$$g(\varepsilon) = \frac{8}{\gamma a^2 (2\pi)^2} \ln \frac{2 q_x a \sqrt{\gamma}}{\sqrt{\varepsilon}} = \frac{8}{\gamma a^2 (2\pi)^2} \left[\ln 2 q_x a \sqrt{\gamma} - \frac{1}{2} \ln \varepsilon \right] = - \frac{1}{\gamma a^2 (2\pi)^2} \ln \varepsilon$$

sedlo v 2D vodi do logaritmske divergence

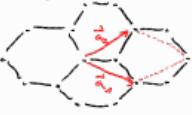
sedlo v 3D vodi do nezveznega odvoda, a brez divergence

$$\xi(\vec{k}) = \xi_{\text{at}} - \frac{\beta + \sum_{\vec{R}} f(\vec{R}) e^{i\vec{k} \cdot \vec{R}}}{1 + \sum_{\vec{R}} \alpha(\vec{R}) e^{i\vec{k} \cdot \vec{R}}}$$

samo za Brav. mreže

Primer, kjer formula ne moremo uporabiti:

grafen: sestoji se



$$\vec{r} = \vec{R} + \vec{r}_n = m_1 \vec{a}_1 + m_2 \vec{a}_2 + \left\{ \vec{r}_1 \right. \\ \left. \vec{r}_2 \right.$$

$$\vec{r}_1 = 0, \vec{r}_2 = \frac{\vec{a}_1 + \vec{a}_2}{3}$$

sp-hibridizacija

$$C: 1s^2 2s^2 2p^2 \\ 2sp^2 2p_z$$

$$H|\psi\rangle = \varepsilon|\psi\rangle$$

$$|\psi\rangle = \sum_{\vec{R}_n} C_{\vec{R}_n} |\vec{R}_n\rangle$$

$$\langle F | \vec{R}_n \rangle = \psi_{p_z}(\vec{r} - \vec{R}_n - \vec{r}_n)$$

$$H \sum_{\vec{R}_n} C_{\vec{R}_n} |\vec{R}_n\rangle = \varepsilon \sum_{\vec{R}_n} C_{\vec{R}_n} |\vec{R}_n\rangle \quad / \quad \langle \vec{R}_n |$$

$$\sum_{\vec{R}_n} C_{\vec{R}_n} \langle \vec{R}_n | H | \vec{R}'_n \rangle = \varepsilon \sum_{\vec{R}_n} C_{\vec{R}_n} \langle \vec{R}_n | \vec{R}'_n \rangle$$

$$S_{\vec{R}_n, \vec{R}'_n} \text{ (skoraj } \underline{I}, \underline{S} = \underline{I} + \underline{\alpha}, \alpha_{ij} \ll 1)$$

$$\downarrow \text{stolpec} \quad \downarrow \text{matrika} \quad \downarrow \text{stolpec} \quad \downarrow \text{matrika}$$

$$\underline{H} \underline{C} = \varepsilon \underline{S} \underline{C} \quad \text{pospl. probl. last. vr.}$$

$$\underline{S}^{-1} \underline{H} \underline{C} = \varepsilon \underline{C} \quad (1+x)^{-1} = 1-x \rightarrow \underline{S}^{-1} \text{ tudi skoraj } \underline{I}$$

$$\rightarrow \underline{S}^{-1} \rightarrow \underline{I} \text{ (zanemarimo)}$$

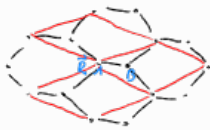
$$\underline{H} \underline{C} = \varepsilon \underline{C}$$

$$\rightarrow \sum_{\vec{R}_n} H_{\vec{R}_n, \vec{R}'_n} C_{\vec{R}'_n} = \varepsilon C_{\vec{R}_n}$$

$$\langle \vec{R}_n | H | \vec{R}'_n \rangle = \langle \vec{R}_n | H_{\vec{R}_n}^{\text{at}} | \vec{R}'_n \rangle + \langle \vec{R}_n | \Delta V | \vec{R}'_n \rangle = \varepsilon^{\text{at}} \langle \vec{R}_n | \vec{R}'_n \rangle + \langle \vec{R}_n | \Delta V | \vec{R}'_n \rangle = \begin{cases} \varepsilon^{\text{at}} - \beta(\vec{R}) & \vec{R}_n = \vec{R}'_n \\ -\gamma & \vec{R}_n \neq \vec{R}'_n, \text{ najbližji sosed } \\ 0 & \vec{R}_n \neq \vec{R}'_n, \text{ niso najbližji sosedi} \end{cases}$$

$$\rightarrow H_{\vec{R}_n, \vec{R}'_n}^{\text{at}} + \Delta V$$

$$\vec{R}_n: (\varepsilon^{\text{at}} - \beta) C_{\vec{R}_n} + \sum_{\vec{R}'_n} (-\gamma) C_{\vec{R}'_n} = \varepsilon C_{\vec{R}_n} \\ \rightarrow \text{najbližji sosedi}$$



$\vec{R}_\alpha, \alpha \in \{A, B\}$

Najbližji sosed $\vec{R}_A$:

- $\vec{R}_B$
- $(\vec{R} + \vec{a}_1)_B$
- $(\vec{R} + \vec{a}_2)_B$

Najbližji sosed $\vec{R}_B$:

- $\vec{R}_A$
- $(\vec{R} + \vec{a}_1)_A$
- $(\vec{R} + \vec{a}_2)_A$

$$\rightarrow (\xi^{A+} - \beta) c_{\vec{R}_A} - \gamma c_{\vec{R}_B} - \gamma c_{(\vec{R} + \vec{a}_1)_B} - \gamma c_{(\vec{R} + \vec{a}_2)_B} = \xi c_{\vec{R}_A}$$

$$(\xi^{B+} - \beta) c_{\vec{R}_B} - \gamma c_{\vec{R}_A} - \gamma c_{(\vec{R} + \vec{a}_1)_A} - \gamma c_{(\vec{R} + \vec{a}_2)_A} = \xi c_{\vec{R}_B}$$

upoštevamo transl. sim.

$$c_{\vec{R}_A} = c_A e^{i\vec{k} \cdot \vec{R}}, \quad c_{\vec{R}_B} = c_B e^{i\vec{k} \cdot \vec{R}}$$

$$(\xi^{A+} - \beta) c_A e^{i\vec{k} \cdot \vec{R}} - \gamma c_B e^{i\vec{k} \cdot \vec{R}} - \gamma c_B e^{i\vec{k} \cdot (\vec{R} + \vec{a}_1)} - \gamma c_B e^{i\vec{k} \cdot (\vec{R} + \vec{a}_2)} = \xi c_A e^{i\vec{k} \cdot \vec{R}}$$

$$(\xi^{A+} - \beta) c_A - \gamma c_B (1 + e^{-i\vec{k} \cdot \vec{a}_1} + e^{-i\vec{k} \cdot \vec{a}_2}) = \xi c_A$$

$$(\xi^{A+} - \beta) c_B - \gamma c_A (1 + e^{i\vec{k} \cdot \vec{a}_1} + e^{i\vec{k} \cdot \vec{a}_2}) = \xi c_B$$

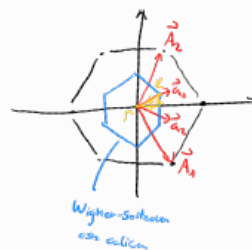
$$\begin{bmatrix} \xi^{A+} - \beta - \xi & -\gamma(1 + e^{-i\vec{k} \cdot \vec{a}_1} + e^{-i\vec{k} \cdot \vec{a}_2}) \\ -\gamma(1 + e^{i\vec{k} \cdot \vec{a}_1} + e^{i\vec{k} \cdot \vec{a}_2}) & \xi^{B+} - \beta - \xi \end{bmatrix} \begin{bmatrix} c_A \\ c_B \end{bmatrix} = 0$$

$$\det \dots = 0 = (\xi^{A+} - \beta - \xi)^2 - \gamma^2 |1 + e^{i\vec{k} \cdot \vec{a}_1} + e^{i\vec{k} \cdot \vec{a}_2}|^2 =$$

$$= (\xi^{A+} - \beta - \xi + \gamma |1 + e^{i\vec{k} \cdot \vec{a}_1} + e^{i\vec{k} \cdot \vec{a}_2}|)(\xi^{A+} - \beta - \xi - \gamma |1 + e^{i\vec{k} \cdot \vec{a}_1} + e^{i\vec{k} \cdot \vec{a}_2}|)$$

$$\rightarrow \xi_{\pm} = \xi^{A+} - \beta \pm \gamma |1 + e^{i\vec{k} \cdot \vec{a}_1} + e^{i\vec{k} \cdot \vec{a}_2}|$$

$$\xi_{\pm} = \pm \gamma |1 + e^{i\vec{k} \cdot \vec{a}_1} + e^{i\vec{k} \cdot \vec{a}_2}|$$



$$k = \frac{1}{3}(\vec{a}_1 + 2\vec{a}_2)$$

$$\xi(k + \vec{q}) = ? \quad |\vec{q}| < \frac{\pi}{a}$$

$$\xi(k + \vec{q}) = \pm \gamma |1 + e^{i\frac{1}{3}(\vec{a}_1 + 2\vec{a}_2 + \vec{q}) \cdot \vec{a}_1} + e^{i\frac{1}{3}(\vec{a}_1 + 2\vec{a}_2 + \vec{q}) \cdot \vec{a}_2}| =$$

$$= \pm \gamma |1 + e^{i(\frac{1}{3} + \vec{q} \cdot \vec{a}_1)} + e^{i(\frac{1}{3} + \vec{q} \cdot \vec{a}_2)}| =$$

$$= \pm \gamma |1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_1) + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_2)| =$$

$$= \pm \gamma |1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_2))| =$$

$$= \pm \gamma |1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_2))| =$$

$$= \pm \gamma |1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_2))| =$$

$$= \pm \gamma |1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_2))| =$$

$$= \pm \gamma |1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_2))| =$$

$$= \pm \gamma |1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_2))| =$$

$$= \pm \gamma |1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_1 + e^{i\frac{1}{3}}(1 + i\vec{q} \cdot \vec{a}_2))| =$$

Kvaziklasichen priblizhek (ciklotronskan efektivna masen)

$$\hbar \dot{\vec{k}} = \vec{F} = e(\vec{E} + \vec{v} \times \vec{B}) \quad \omega_c = \frac{eB}{\hbar}$$

$$\vec{v} = \frac{1}{\hbar} \frac{\partial \xi(\vec{k})}{\partial \vec{k}} = \frac{1}{\hbar} \frac{\partial \xi}{\partial \vec{k}} \quad \nabla_{\vec{k}} \xi = \hbar \vec{v}$$

$$\xi(\vec{k}) = \sum_{n, i} \frac{\hbar^2}{2} \vec{k} \cdot \vec{m}_i \vec{k}$$

$$\nabla_{\vec{k}} \xi = \hbar \vec{v}$$

$$\hbar \dot{\vec{k}} = \nabla_{\vec{k}} \xi = e(\vec{E} + \vec{v} \times \vec{B}), \quad \vec{v} = \vec{v}_0 e^{-i\omega t}$$

$$-i\omega \nabla_{\vec{k}} \xi = e \vec{v}_0 \times \vec{B}$$

$$\nabla_{\vec{k}} \xi = i \frac{e}{\omega} \vec{v}_0 \times \vec{B}, \quad \vec{B} = B \vec{e}_z$$

$$\begin{pmatrix} m_{xx} & m_{xy} & m_{xz} \\ m_{yx} & m_{yy} & m_{yz} \\ m_{zx} & m_{zy} & m_{zz} \end{pmatrix} \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = i \frac{e}{\omega} \begin{pmatrix} v_y B \\ -v_x B \\ 0 \end{pmatrix}$$

→ tenzor lahko simetriziramo (mim. sta v_y in v_x vzamemo $\frac{1}{2}(v_y + v_x)$)

$$\begin{pmatrix} m_{xx} & m_{xy} - i \frac{e}{\omega} B & m_{xz} \\ m_{yx} + i \frac{e}{\omega} B & m_{yy} & m_{yz} \\ m_{zx} & m_{zy} & m_{zz} \end{pmatrix} \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = 0$$

$$\det(\dots) = 0 = m_{xx} m_{yy} m_{zz} - m_{xz}^2 - (m_{xy} - i \frac{e}{\omega} B)(m_{yx} + i \frac{e}{\omega} B) m_{yz} + (m_{xy} - i \frac{e}{\omega} B) m_{yz} m_{xz} + m_{xz} (m_{yx} + i \frac{e}{\omega} B) m_{yz} - m_{xz}^2 m_{zz} = \det(\underline{m}) - m_{zz}^2 (\frac{e}{\omega} B)^2$$

$$\rightarrow \omega = \frac{eB}{\hbar} \sqrt{\frac{m_{zz}}{\det(\underline{m})}} = \frac{eB}{\hbar} \frac{1}{m_c^*}, \quad m_c^* = \sqrt{\frac{\det(\underline{m})}{m_{zz}}} = \sqrt{\frac{\det(\underline{m})}{\vec{e}_z \cdot \underline{m} \cdot \vec{e}_z}}, \quad \vec{e}_z = \frac{\vec{B}}{|B|}$$

analogija iz kl. fiz. → efektivna ciklotronska masa → neod. od izbr. koord. sist.